

- 6 (a) (i)** Resistance of the bulb is defined as the ratio $\frac{V}{I}$ where V is the potential difference across the bulb and I is the current flowing in it. [B1]

Comments: It is best to provide the expression for resistance, rather than simply answering that it is the ratio of potential difference to current. Resistance is not potential difference per unit current.

$$(ii) \quad P = \frac{V^2}{R}$$

$$60 = \frac{(240)^2}{R} \quad [C1]$$

$$R = 960 \, \Omega \quad [A1]$$

$$(b) \quad R = \frac{\rho L}{A} = \frac{\rho L}{\left(\frac{\pi d^2}{4}\right)}$$

$$960 = \frac{(7.9 \times 10^{-7})(0.14)}{\left(\frac{\pi d^2}{4}\right)} \quad [C1]$$

$$d = 1.2 \times 10^{-5} \, \text{m} = 0.012 \, \text{mm} \quad [A1]$$

- (c)** The ratio of length to cross-section area is the same for all the filaments. [B1]

- (d) (i)** Filament C is used to make the bulb with higher temperature. [A1]

- (ii)** Filament A is used to make the bulb that can last 2000 hours. [A1]

- (e)** Missing values in italics, from 3rd to 5th row, one mark for each correct row. [A3]

	A	B	C
resistance/ Ω	960	960	960
diameter/ mm	0.0129	0.012	0.0113
length/ m	0.159	0.14	0.122
surface area/ m^2	6.4×10^{-6}	5.3×10^{-6}	4.3×10^{-6}
power lost per unit area/ W m^{-2}	0.93×10^7	1.13×10^7	1.39×10^7

Comments: Surface area refers to the curved area and the cross-sectional areas of the wire.

- (f) From the graph,
 filament A temperature = 2420 °C [A1]
 filament B temperature = 2600 °C [A1]
 filament C temperature = 2730 °C [A1]

Comments: There were some careless reading from the graph. It is expected that readings should be to half the smallest division, similar to practical.

- (g) (i) For lowest temperature bulb,
 running cost = cost of using bulb + cost of replacing bulb
 = (0.060 kW)(2000 h)(\$0.172 kWh^{-1}) + \$0.50
 = \$21.14 [B1]

For highest temperature bulb ,
 running cost = cost of using bulb + cost of replacing bulb
 = (0.060 kW)(2000 h)(\$0.172 kWh^{-1}) + \$2.00
 = \$22.64 [B1]

Hence bulb with highest temperature has a higher running cost. [A1]

- (ii) light energy from lowest temperature bulb for 2000 hours
 = (0.090)(60) (2000 $\times$ 3600) = 3.9×10^7 J [B1]

light energy from highest temperature bulb for 2000 hours
 = (0.12)(60)(2000 $\times$ 3600) = 5.2×10^7 J [B1]

$$\frac{\text{cost of a unit of light energy from lowest temperature bulb}}{\text{cost of a unit of light energy from highest temperature bulb}} = \frac{21.14/3.9}{22.64/5.2} \quad [\text{M1}]$$

$$= 1.2 \quad [\text{A1}]$$

Comments: This part is poorly done. Many students do not know how to use the given electricity cost \$0.172 kWh^{-1} in their calculation.

- (iii) (Compact) fluorescent lightbulbs and LEDs [B1]
 (iv) Higher energy efficiency/ longer lifespan [B1]

Comments: saves energy or uses less energy/power or brighter light are not acceptable as they are too generic and need to be elaborated further.